

Qualification Pack



Industrial IoT and Edge AI Analyst

QP Code: ITG/ELE/Q0101

Version: 1.0

NSQF Level: 5.5

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ITG/ELE/Q0101: Industrial IoT and Edge AI Analyst

Brief Job Description

Learners will understand the prevailing difficulties for Industries to adopt IoT, Learner will be able to design and develop an intelligent and sustainable digital transformation solution under Industry 4.0/Industry 5.0 with cloud server and all needed securities.

Personal Attributes

Knowledge of programing (preferably C/C++, python) and Basic Electronics and Embedded System

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [ITG/ELE/N0101: Advance Embedded C](#)
2. [ITG/ELE/N0102: ARM based Multicore Architecture](#)
3. [ITG/ELE/N0103: Embedded Linux](#)
4. [ITG/ELE/N0104: IoT Architecture and Deployment](#)
5. [ITG/ELE/N0105: Industrial IoT](#)
6. [ITG/ELE/N0107: Edge AI](#)
7. [DGT/VSQ/N0103: Employability Skills \(90 Hours\)](#)
8. [ITG/ELE/N0106: Cloud Management for IoT](#)

Qualification Pack (QP) Parameters

Sector	Electronics
Sub-Sector	Strategic Electronics
Occupation	Industrial IoT and Edge AI Analyst
Country	India
NSQF Level	5.5



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Credits	25
Aligned to NCO/ISCO/ISIC Code	2152, 2511
Minimum Educational Qualification & Experience	Completed 4 year UG (in case of 4-year UG with honours/ honours with research) (IT, electronics, Electrical, Computer Science) with 2 Years of experience Diploma OR Diploma (IT, electronics, Electrical, Computer Science) with 2 Years of experience Diploma
Minimum Level of Education for Training in School	
Pre-Requisite License or Training	NA
Minimum Job Entry Age	20 Years
Last Reviewed On	NA
Next Review Date	31/08/2026
NSQC Approval Date	19/09/2024
Version	1.0
Reference code on NQR	NCVET-QG-5.5-EH-00736-2023-V1-IITG
NQR Version	1



Qualification Pack

ITG/ELE/N0101: Advance Embedded C

Description

Advance Embedded C

Scope

The scope covers the following :

- Advance Embedded C

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Working knowledge of Advance Embedded C programming and its salient features for embedded systems
- PC2.** Able to Develop Embedded C code for Complex Embedded Products
- PC3.** Able to test and debug the code on Embedded platforms
- PC4.** Able to write optimized code as per given Memory and MIPS resources
- PC5.** Project

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Theoretical and practical knowledge in implementation of real time embedded, IoT and smart systems and Edge AI

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** A Professional must be able to tackle all the issues related to complex Embedded, IOT, Edge AI System, and should be able to design a cost effective solution for the same



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	20	60	15	5
PC1. Working knowledge of Advance Embedded C programming and its salient features for embedded systems	4	10	-	-
PC2. Able to Develop Embedded C code for Complex Embedded Products	4	10	-	-
PC3. Able to test and debug the code on Embedded platforms	4	20	-	-
PC4. Able to write optimized code as per given Memory and MIPS resources	4	20	-	-
PC5. Project	4	-	15	5
NOS Total	20	60	15	5



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National Occupational Standards (NOS) Parameters

NOS Code	ITG/ELE/N0101
NOS Name	Advance Embedded C
Sector	Electronics
Sub-Sector	
Occupation	Industrial IoT and Edge AI Analyst
NSQF Level	5.5
Credits	2
Version	1.0
Last Reviewed Date	19/09/2024
Next Review Date	31/08/2026
NSQC Clearance Date	19/09/2024



Qualification Pack

ITG/ELE/N0102: ARM based Multicore Architecture

Description

ARM based Multicore Architecture

Scope

The scope covers the following :

- ARM based Multicore Architecture

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

PC1. Ability to define the features of embedded systems, architecture of ARM and applications.

PC2. Analyze and understand the instruction set and development tools of ARM

PC3. Analyze and understand the THUMB state and achieving competency in assembly programming of ARM.

PC4. • Ability to identify and use of the exception, interrupts and interrupt
• handling schemes

PC5. Ability to design with the Multicore Architectural features.

PC6. Ability to leverage the Memory management unit given by the used platform

PC7. Ability to profile MIPS and power consumed

PC8. Able to Define RISC vs RISC-V Architecture

PC9. Project.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Theoretical and practical knowledge in implementation of real time embedded, IoT and smart systems and Edge AI

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. A Professional must be able to tackle all the issues related to complex Embedded, IOT, Edge AI System, and should be able to design a cost effective solution for the same

GS2. Various processes in complex Embedded, IOT, Edge AI System Technology and a professional will have knowledge of various procedures related to Embedded system technology



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	20	60	15	5
PC1. Ability to define the features of embedded systems, architecture of ARM and applications.	2	5	-	-
PC2. Analyze and understand the instruction set and development tools of ARM	2	5	-	-
PC3. Analyze and understand the THUMB state and achieving competency in assembly programming of ARM.	2	5	-	-
PC4. • Ability to identify and use of the exception, interrupts and interrupt • handling schemes	2	5	-	-
PC5. Ability to design with the Multicore Architectural features.	2	10	-	-
PC6. Ability to leverage the Memory management unit given by the used platform	2	10	-	-
PC7. Ability to profile MIPS and power consumed	2	10	-	-
PC8. Able to Define RISC vs RISC-V Architecture	2	10	-	-
PC9. Project.	4	-	15	5
NOS Total	20	60	15	5



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National Occupational Standards (NOS) Parameters

NOS Code	ITG/ELE/N0102
NOS Name	ARM based Multicore Architecture
Sector	Electronics
Sub-Sector	
Occupation	Industrial IoT and Edge AI Analyst
NSQF Level	5.5
Credits	2
Version	1.0
Last Reviewed Date	19/09/2024
Next Review Date	31/08/2026
NSQC Clearance Date	19/09/2024



Qualification Pack

ITG/ELE/N0103: Embedded Linux

Description

Embedded Linux

Scope

The scope covers the following :

- Embedded Linux

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Able to use right variant of Linux and build it properly.
- PC2.** Ability to write device drivers for various peripherals
- PC3.** Ability to write code to communicate with various peripherals.
- PC4.** Ability to utilize all resources provided by platform/system to have the best design.
- PC5.** Ability to run Debug and performance test.
- PC6.** Able to port Embedded Linux on any capable SoC/SoM.
- PC7.**
 - Able to work with Process management, Synchronization among processes, IPC, socket programming etc
- PC8.** Ability to make bootloader and do the required porting.
- PC9.** Able to create a Yocto build setup.
- PC10.** Project

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Theoretical and practical knowledge in implementation of real time embedded, IoT and smart systems and Edge AI

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** A Professional must be able to tackle all the issues related to complex Embedded, IOT, Edge AI System, and should be able to design a cost effective solution for the same
- GS2.** Reasonably good in Various processes in complex Embedded, IOT, Edge AI System Technology. A professional will have knowledge of various procedures related to Embedded system technology



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	20	60	15	5
PC1. Able to use right variant of Linux and build it properly.	2	5	-	-
PC2. Ability to write device drivers for various peripherals	2	5	-	-
PC3. Ability to write code to communicate with various peripherals.	2	5	-	-
PC4. Ability to utilize all resources provided by platform/system to have the best design.	2	5	-	-
PC5. Ability to run Debug and performance test.	2	5	-	-
PC6. Able to port Embedded Linux on any capable SoC/SoM.	2	10	-	-
PC7. • Able to work with Process management, Synchronization among • processes, IPC, socket programming etc	2	10	-	-
PC8. Ability to make bootloader and do the required porting.	2	10	-	-
PC9. Able to create a Yocto build setup.	2	5	-	-
PC10. Project	2	-	15	5
NOS Total	20	60	15	5



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National Occupational Standards (NOS) Parameters

NOS Code	ITG/ELE/N0103
NOS Name	Embedded Linux
Sector	Electronics
Sub-Sector	
Occupation	Industrial IoT and Edge AI Analyst, Industrial IoT and Edge AI Analyst
NSQF Level	5.5
Credits	4
Version	1.0
Last Reviewed Date	19/09/2024
Next Review Date	31/08/2026
NSQC Clearance Date	19/09/2024



Qualification Pack

ITG/ELE/N0104: IoT Architecture and Deployment

Description

IoT Architecture and Deployment

Scope

The scope covers the following :

- IoT Architecture and Deployment

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Ability to use IoT feature while designing any application
- PC2.** Ability to make design based on given constraint of the device
- PC3.** Ability to utilize the IOT Protocols (MQTT/COAP/HTTP) and use to get the best system performance
- PC4.** Ability to design Low power systems
- PC5.** Ability to design end to end solution including visualization e.g. Node-Red.
- PC6.** Ability to work with wireless sensor networks and Architectures
- PC7.** Ability to use and debug Networking Sensors
- PC8.** Ability to identify IoT Deployment challenges
- PC9.** Ability to work on LPWAN Systems
- PC10.** Project

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Theoretical and practical knowledge in implementation of real time embedded, IoT and smart systems and Edge AI

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** • A Professional must be able to tackle all the issues related to complex Embedded, IOT, Edge AI System, and should be able to design a cost effective solution for the same
- GS2.** Reasonably good in: Various processes in complex Embedded, IOT, Edge AI System Technology and a professional will have knowledge of various procedures related to Embedded system technology



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	20	60	15	5
PC1. Ability to use IoT feature while designing any application	2	5	-	-
PC2. Ability to make design based on given constraint of the device	2	5	-	-
PC3. Ability to utilize the IOT Protocols (MQTT/COAP/HTTP) and use to get the best system performance	2	5	-	-
PC4. Ability to design Low power systems	2	5	-	-
PC5. Ability to design end to end solution including visualization e.g. Node-Red.	2	5	-	-
PC6. Ability to work with wireless sensor networks and Architectures	2	10	-	-
PC7. Ability to use and debug Networking Sensors	2	10	-	-
PC8. Ability to identify IoT Deployment challenges	2	10	-	-
PC9. Ability to work on LPWAN Systems	2	5	-	-
PC10. Project	2	-	15	5
NOS Total	20	60	15	5



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National Occupational Standards (NOS) Parameters

NOS Code	ITG/ELE/N0104
NOS Name	IoT Architecture and Deployment
Sector	Electronics
Sub-Sector	
Occupation	Industrial IoT and Edge AI Analyst
NSQF Level	5.5
Credits	4
Version	1.0
Last Reviewed Date	19/09/2024
Next Review Date	31/08/2026
NSQC Clearance Date	19/09/2024



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ITG/ELE/N0105: Industrial IoT

Description

Industrial IoT

Scope

The scope covers the following :

- Industrial IoT

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Understand the Industrial Automation and IIoT Architecture
- PC2.** Overview of oneM2M
- PC3.** Analyze and understand IoT Ecosystem-PLC/SCAD, OPC-UA
- PC4.** Understand Modbus (RS485), Profibus, Ethernet, Ethercat
- PC5.** Understand Data Acquisition
- PC6.** Overview of Data Visualization
- PC7.** Learn about Data analytics and Rule engine
- PC8.** Understand Business Intelligence
- PC9.** Learn about OEE Dashboard
- PC10.** Project

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Theoretical and practical knowledge in implementation of real time embedded, IoT and smart systems and Edge AI

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** • A Professional must be able to tackle all the issues related to complex Embedded, IOT, Edge AI System, and should be able to design a cost effective solution for the same
- GS2.** Reasonably good in: Various processes in complex Embedded, IOT, Edge AI System Technology and a professional will have knowledge of various procedures related to Embedded system technology



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	20	60	15	5
PC1. Understand the Industrial Automation and IIoT Architecture	2	5	-	-
PC2. Overview of oneM2M	2	5	-	-
PC3. Analyze and understand IoT Ecosystem- PLC/SCAD, OPC-UA	2	5	-	-
PC4. Understand Modbus (RS485), Profibus, Ethernet, Ethercat	2	5	-	-
PC5. Understand Data Acquisition	2	5	-	-
PC6. Overview of Data Visualization	2	10	-	-
PC7. Learn about Data analytics and Rule engine	2	10	-	-
PC8. Understand Business Intelligence	2	10	-	-
PC9. Learn about OEE Dashboard	2	5	-	-
PC10. Project	2	-	15	5
NOS Total	20	60	15	5



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National Occupational Standards (NOS) Parameters

NOS Code	ITG/ELE/N0105
NOS Name	Industrial IoT
Sector	Electronics
Sub-Sector	
Occupation	Industrial IoT and Edge AI Analyst
NSQF Level	5.5
Credits	4
Version	1.0
Last Reviewed Date	19/09/2024
Next Review Date	31/08/2026
NSQC Clearance Date	19/09/2024



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ITG/ELE/N0107: Edge AI

Description

Edge AI

Scope

The scope covers the following :

- Edge AI

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Ability to use the CPU, GPU, TPU Architecture
- PC2.** Ability to define Constraints of Edge Devices
- PC3.** Understand Machine Learning
- PC4.** Ability to deploy Inferencing
- PC5.** Write programming for Deployment of ML.
- PC6.** Design Edge computing based solutions
- PC7.** Ability to define the constraints of ML
- PC8.** Performance assessment on Edge devices.
- PC9.** Ability to use TinyML, Edge Impulse, Tensorflow Lite
- PC10.** Project

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Theoretical and practical knowledge in implementation of real time embedded, IoT and smart systems and Edge AI

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** A Professional must be able to tackle all the issues related to complex Embedded, IOT, Edge AI System, and should be able to design a cost effective solution for the same
- GS2.** Reasonably good in: Various processes in complex Embedded, IOT, Edge AI System Technology. A professional will have knowledge of various procedures related to Embedded system technology



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	20	60	15	5
PC1. Ability to use the CPU, GPU, TPU Architecture	2	5	-	-
PC2. Ability to define Constraints of Edge Devices	2	5	-	-
PC3. Understand Machine Learning	2	5	-	-
PC4. Ability to deploy Inferencing	2	5	-	-
PC5. Write programming for Deployment of ML.	2	5	-	-
PC6. Design Edge computing based solutions	2	10	-	-
PC7. Ability to define the constraints of ML	2	10	-	-
PC8. Performance assessment on Edge devices.	2	10	-	-
PC9. Ability to use TinyML, Edge Impulse, Tensorflow Lite	2	5	-	-
PC10. Project	2	-	15	5
NOS Total	20	60	15	5



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National Occupational Standards (NOS) Parameters

NOS Code	ITG/ELE/N0107
NOS Name	Edge AI
Sector	Electronics
Sub-Sector	
Occupation	Industrial IoT and Edge AI Analyst, Industrial IoT and Edge AI Analyst
NSQF Level	5.5
Credits	4
Version	1.0
Last Reviewed Date	19/09/2024
Next Review Date	31/08/2026
NSQC Clearance Date	19/09/2024



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DGT/VSQ/N0103: Employability Skills (90 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** understand the significance of employability skills in meeting the current job market requirement and future of work
- PC2.** identify and explore learning and employability relevant portals
- PC3.** research about the different industries, job market trends, latest skills required and the available opportunities

Constitutional values - Citizenship

To be competent, the user/individual on the job must be able to:

- PC4.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC5.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC6.** recognize the significance of 21st Century Skills for employment



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- PC7.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life
- PC8.** adopt a continuous learning mindset for personal and professional development

Basic English Skills

To be competent, the user/individual on the job must be able to:

- PC9.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC10.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC11.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC12.** identify career goals based on the skills, interests, knowledge, and personal attributes
- PC13.** prepare a career development plan with short- and long-term goals

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC14.** follow verbal and non-verbal communication etiquette while communicating in professional and public settings
- PC15.** use active listening techniques for effective communication
- PC16.** communicate in writing using appropriate style and format based on formal or informal requirements
- PC17.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC18.** communicate and behave appropriately with all genders and PwD
- PC19.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC20.** identify and select reliable institutions for various financial products and services such as bank account, debit and credit cards, loans, insurance etc.
- PC21.** carry out offline and online financial transactions, safely and securely, using various methods and check the entries in the passbook
- PC22.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC23.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC24.** operate digital devices and use their features and applications securely and safely
- PC25.** carry out basic internet operations by connecting to the internet safely and securely, using the mobile data or other available networks through Bluetooth, Wi-Fi, etc.
- PC26.** display responsible online behaviour while using various social media platforms



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- PC27.** create a personal email account, send and process received messages as per requirement
- PC28.** carry out basic procedures in documents, spreadsheets and presentations using respective and appropriate applications
- PC29.** utilize virtual collaboration tools to work effectively

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC30.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC31.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC32.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC33.** identify different types of customers and ways to communicate with them
- PC34.** identify and respond to customer requests and needs in a professional manner
- PC35.** use appropriate tools to collect customer feedback
- PC36.** follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

- PC37.** create a professional Curriculum vitae (Résumé)
- PC38.** search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively
- PC39.** apply to identified job openings using offline /online methods as per requirement
- PC40.** answer questions politely, with clarity and confidence, during recruitment and selection
- PC41.** identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** need for employability skills and different learning and employability related portals
- KU2.** various constitutional and personal values
- KU3.** different environmentally sustainable practices and their importance
- KU4.** Twenty first (21st) century skills and their importance
- KU5.** how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up
- KU6.** importance of career development and setting long- and short-term goals
- KU7.** about effective communication
- KU8.** POSH Act
- KU9.** Gender sensitivity and inclusivity
- KU10.** different types of financial institutes, products, and services



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- KU11.** components of salary and how to compute income and expenditure
- KU12.** importance of maintaining safety and security in offline and online financial transactions
- KU13.** different legal rights and laws
- KU14.** different types of digital devices and the procedure to operate them safely and securely
- KU15.** how to create and operate an e- mail account
- KU16.** use applications such as word processors, spreadsheets etc.
- KU17.** how to identify business opportunities
- KU18.** types and needs of customers
- KU19.** how to apply for a job and prepare for an interview
- KU20.** apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read and write different types of documents/instructions/correspondence in English and other languages
- GS2.** communicate effectively using appropriate language in formal and informal settings
- GS3.** behave politely and appropriately with all to maintain effective work relationship
- GS4.** how to work in a virtual mode, using various technological platforms
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. understand the significance of employability skills in meeting the current job market requirement and future of work	-	-	-	-
PC2. identify and explore learning and employability relevant portals	-	-	-	-
PC3. research about the different industries, job market trends, latest skills required and the available opportunities	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC4. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC5. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	1	3	-	-
PC6. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC7. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
PC8. adopt a continuous learning mindset for personal and professional development	-	-	-	-
<i>Basic English Skills</i>	3	4	-	-
PC9. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC11. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-
PC12. identify career goals based on the skills, interests, knowledge, and personal attributes	-	-	-	-
PC13. prepare a career development plan with short- and long-term goals	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC14. follow verbal and non-verbal communication etiquette while communicating in professional and public settings	-	-	-	-
PC15. use active listening techniques for effective communication	-	-	-	-
PC16. communicate in writing using appropriate style and format based on formal or informal requirements	-	-	-	-
PC17. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	1	-	-
PC18. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC19. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC20. identify and select reliable institutions for various financial products and services such as bank account, debit and credit cards, loans, insurance etc.	-	-	-	-
PC21. carry out offline and online financial transactions, safely and securely, using various methods and check the entries in the passbook	-	-	-	-



Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC22. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC23. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	5	-	-
PC24. operate digital devices and use their features and applications securely and safely	-	-	-	-
PC25. carry out basic internet operations by connecting to the internet safely and securely, using the mobile data or other available networks through Bluetooth, Wi-Fi, etc.	-	-	-	-
PC26. display responsible online behaviour while using various social media platforms	-	-	-	-
PC27. create a personal email account, send and process received messages as per requirement	-	-	-	-
PC28. carry out basic procedures in documents, spreadsheets and presentations using respective and appropriate applications	-	-	-	-
PC29. utilize virtual collaboration tools to work effectively	-	-	-	-
<i>Entrepreneurship</i>	2	3	-	-
PC30. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC31. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC32. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC33. identify different types of customers and ways to communicate with them	-	-	-	-



Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC34. identify and respond to customer requests and needs in a professional manner	-	-	-	-
PC35. use appropriate tools to collect customer feedback	-	-	-	-
PC36. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC37. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC38. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC39. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC40. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC41. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0103
NOS Name	Employability Skills (90 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	5
Credits	3
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025



Qualification Pack

ITG/ELE/N0106: Cloud Management for IoT

Description

Cloud Management for IoT

Scope

The scope covers the following :

- Cloud Management for IoT

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Ability to use the features of Cloud and Cloud Computing.
- PC2.** Analyze and understand the AWS/AZURE
- PC3.** Analyze and understand the security in IoT application.
- PC4.** Understand the mbed TLS and SSL
- PC5.** Ability to use the hardware accelerated AES encryption on ARM MCU/MPU.
- PC6.** Designing of IoT Applications
- PC7.** Learn about Deployment on Cloud using EC
- PC8.** Ability to set up, MQTT Broker, NodeRed
- PC9.** Ability to handle Sensor Network Platforms and Tools
- PC10.** Project

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Theoretical and practical knowledge in implementation of real time embedded, IoT and smart systems and Edge AI

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** A Professional must be able to tackle all the issues related to complex Embedded, IOT, Edge AI System, and should be able to design a cost effective solution for the same
- GS2.** Reasonably good in: Various processes in complex Embedded, IOT, Edge AI System Technology. A professional will have knowledge of various procedures related to Embedded system technology.



Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	20	60	15	5
PC1. Ability to use the features of Cloud and Cloud Computing.	2	5	-	-
PC2. Analyze and understand the AWS/AZURE	2	5	-	-
PC3. Analyze and understand the security in IoT application.	2	5	-	-
PC4. Understand the mbed TLS and SSL	2	5	-	-
PC5. Ability to use the hardware accelerated AES encryption on ARM MCU/MPU.	2	5	-	-
PC6. Designing of IoT Applications	2	10	-	-
PC7. Learn about Deployment on Cloud using EC	2	10	-	-
PC8. Ability to set up, MQTT Broker, NodeRed	2	10	-	-
PC9. Ability to handle Sensor Network Platforms and Tools	2	5	-	-
PC10. Project	2	-	15	5
NOS Total	20	60	15	5



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	ITG/ELE/N0106
NOS Name	Cloud Management for IoT
Sector	Electronics
Sub-Sector	
Occupation	Industrial IoT and Edge AI Analyst, Industrial IoT and Edge AI Analyst
NSQF Level	5.5
Credits	2
Version	1.0
Last Reviewed Date	19/09/2024
Next Review Date	31/08/2026
NSQC Clearance Date	19/09/2024

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

- Batches assigned to the assessment agencies for conducting the assessment on SIP or email
- Assessment agencies send the assessment confirmation to VTP/TC looping IITG
- Assessment agency deploys the ToA certified Assessor for executing the assessment
- IITG monitors the assessment process & records

Minimum Aggregate Passing % at QP Level : 50

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Minimum Passing % at NOS Level: 50

(Please note: A Trainee must score the minimum percentage for each NOS separately as well as on the QP as a whole.)



Qualification Pack

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
ITG/ELE/N0101.Advance Embedded C	20	60	15	5	100	13
ITG/ELE/N0102.ARM based Multicore Architecture	20	60	15	5	100	13
ITG/ELE/N0103.Embedded Linux	20	60	15	5	100	13
ITG/ELE/N0104.IoT Architecture and Deployment	20	60	15	5	100	13
ITG/ELE/N0105.Industrial IoT	20	60	15	5	100	13
ITG/ELE/N0107.Edge AI	20	60	15	5	100	13
DGT/VSQ/N0103.Employability Skills (90 Hours)	20	30	-	-	50	9
ITG/ELE/N0106.Cloud Management for IoT	20	60	15	5	100	13
Total	160	450	105	35	750	100



Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training
AA	Assessment Agency
AB	Awarding Body
ISCO	International Standard Classification of Occupations
NCO	National Classification of Occupations
NCrF	National Credit Framework
NOS	National Occupational Standard(s)
NQR	National Qualification Register
NSQF	National Skills Qualifications Framework
OJT	On the Job Training
AA	Assessment Agency
AB	Awarding Body
ISCO	International Standard Classification of Occupations
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Qualification Pack

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OJT	On the Job Training



Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.
Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.



Qualification Pack

Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.
National Occupational Standard	NOS define the measurable performance outcomes required from an individual engaged in a particular task. They list down what an individual performing that task should know and also do.
Qualification	A formal outcome of an assessment and validation process which is obtained when a competent body determines that an individual has achieved learning outcomes to given standards
Qualification File	A Qualification File is a template designed to capture necessary information of a Qualification from the perspective of NSQF compliance. The Qualification File will be normally submitted by the awarding body for the qualification
Sector	A grouping of professional activities on the basis of their main economic function, product, service or technology.
Long Term Training	Long-term skilling means any vocational training program undertaken for a year and above.
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